

Applied Forecasting Methods

WESAD Dataset Stress Prediction Pipeline

High-Resolution Stress Forecasting via Statistical, ML, and DL Models

Group Members

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Reactive vs. Proactive

Most wearables only detect stress *after* it happens.

- ▶ **Goal:** Predict future stress intensity.
- ▶ **Benefit:** Allow users to intervene *before* the peak.
- ▶ **Data:** Multimodal signals (EDA, HR, TEMP).

The Forecasting Objective

Build a mathematically valid pipeline:

- ▶
- ▶
- ▶
- ▶

Dataset: WESAD (Chest-Worn Sensors)

Primary Signals

- ▶ **EDA:** Electrodermal Activity (Skin Conductance).
- ▶ **HR:** Heart Rate (derived from ECG).
- ▶ **TEMP:** Chest Temperature.

Target: Next-observed continuous stress score.

High-Resolution Processed Data (First 5 Rows)

Sec	HR	EDA	TEMP	Stress
0	77.7	5.24	30.1	0.69
1	77.8	5.47	30.1	0.71
2	80.9	6.27	30.1	0.79
3	71.5	6.41	30.1	0.76
4	66.8	6.20	30.1	0.72

Fixing Mistake #1: The Binary Box Problem

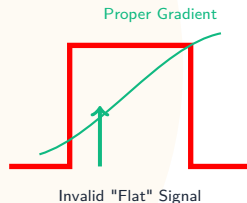
The Issue:

Discrete labels (Stress=1, Baseline=0) create "rectangular" signals.

- ▶ Zero gradient for long periods.
- ▶ Models like ARMA fail to converge.

The Solution:

Engineered a ****Continuous Proxy**** (0.0-1.0) using rolling density to represent physiological stress intensity.



The Mistake

Using $HR(t)$ to predict $Stress(t)$.

This is concurrent detection, not forecasting. The accuracy was artificially high.

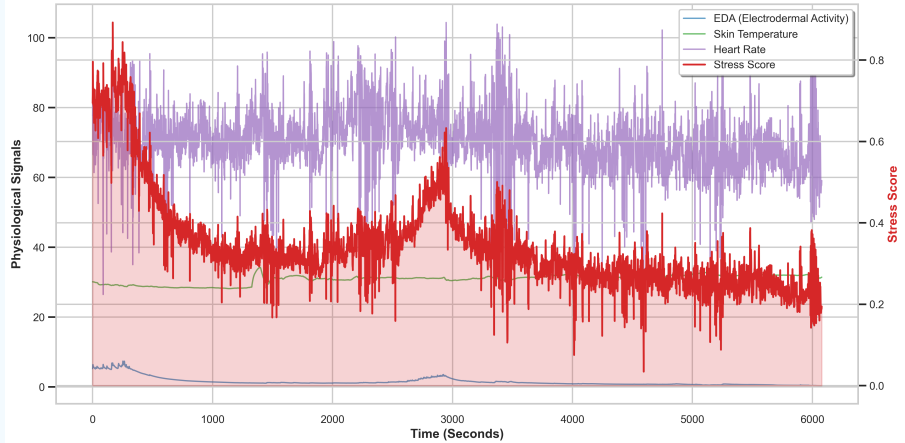
FIX

The Policy

Strict $t - 1$ Lag Strategy

Every sensor input was shifted. To predict now (t), the model only sees the past ($t - 1$).

Subject S2: High-Resolution Continuous Stress Profile

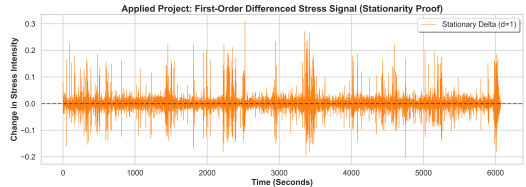


Transformation:

$$\Delta y_t = y_t - y_{t-1}$$

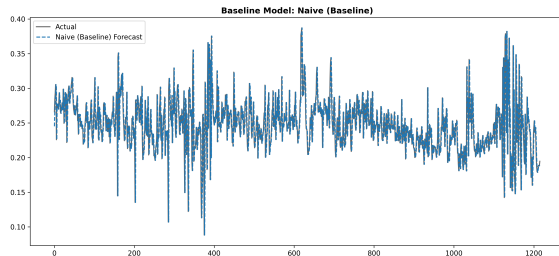
- ▶ **Raw p -value:** 0.066 (Non-Stationary).
- ▶ **Diff p -value:** 0.000 (Stationary).

This stabilized the mean and variance for all statistical models.



THE FORECASTING ENGINE

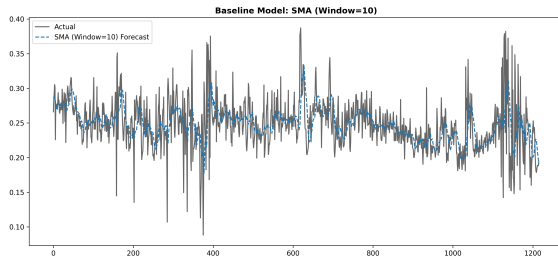
Model 1: Naive Persistence (The baseline)



Logic: $\hat{y}_{t+1} = y_t$

Metric	Value
Model Name	Naive
MAE	0.0211
RMSE	0.0345
MAPE	8.68%
R ² Score	0.0673

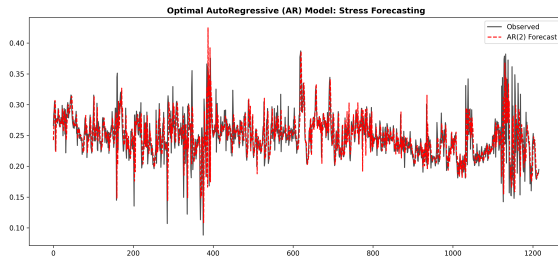
Model 2: Simple Moving Average (SMA-10)



Logic: 10s Window Mean

Metric	Value
Model Name	SMA-10
MAE	0.0229
RMSE	0.0326
MAPE	9.62%
R^2 Score	0.1716

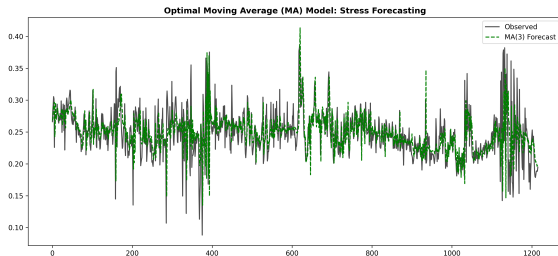
Model 3: AutoRegressive (AR-2)



Lag Choice: PACF lag 2

Metric	Value
Model Name	AR(2)
MAE	0.0210
RMSE	0.0342
MAPE	8.66%
R ² Score	0.0840

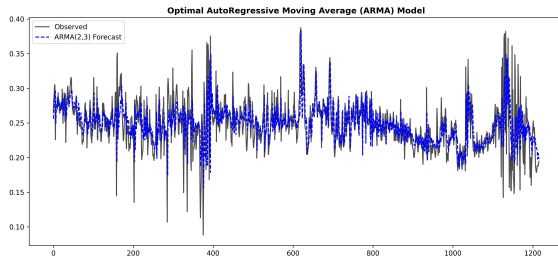
Model 4: Moving Average (MA-3)



Lag Choice: ACF lag 3

Metric	Value
Model Name	MA(3)
MAE	0.0209
RMSE	0.0335
MAPE	8.72%
R ² Score	0.1236

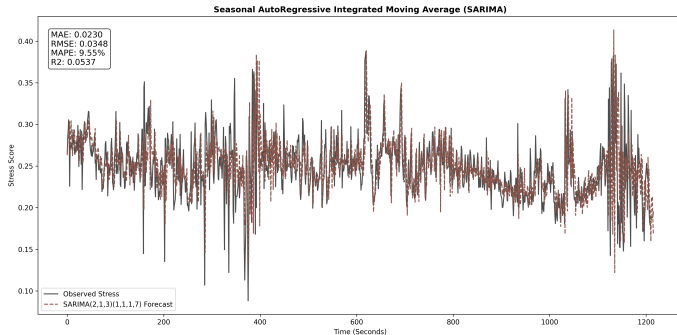
Model 5: ARMA(2,3)



Optimization: AIC Grid Search

Metric	Value
Model Name	ARMA(2,3)
MAE	0.0206
RMSE	0.0316
MAPE	8.60%
R ² Score	0.2204

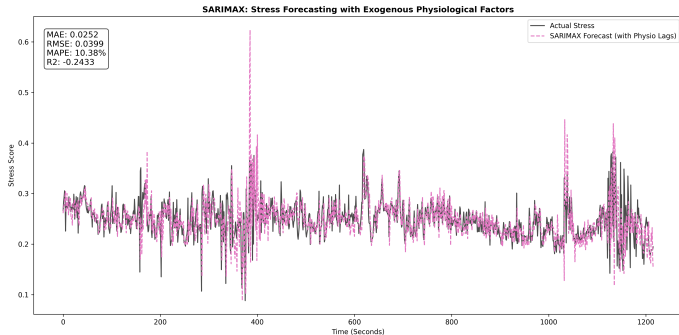
Model 6: Seasonal ARIMA (SARIMA)



Seasonality: 7s Period

Metric	Value
Model Name	SARIMA
MAE	0.0230
RMSE	0.0348
MAPE	9.55%
R ² Score	0.0537

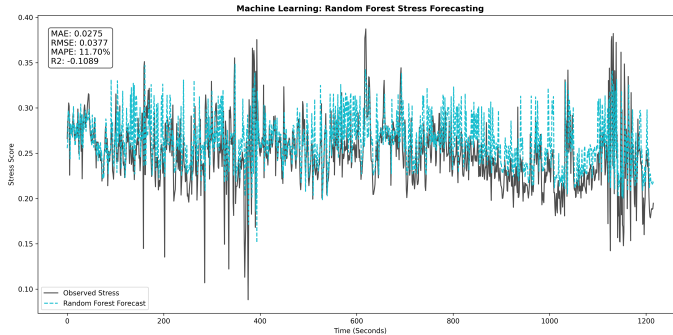
Model 7: SARIMAX (Exogenous Lags)



Features:
HR/EDA/TEMP

Metric	Value
Model Name	SARIMAX
MAE	0.0252
RMSE	0.0399
MAPE	10.38%
R ² Score	-0.2433

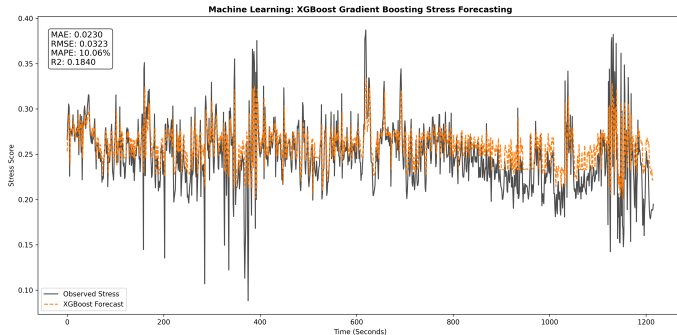
Model 8: Random Forest Ensemble



Type: 100 Trees

Metric	Value
Model Name	RF
MAE	0.0275
RMSE	0.0377
MAPE	11.70%
R ² Score	-0.1089

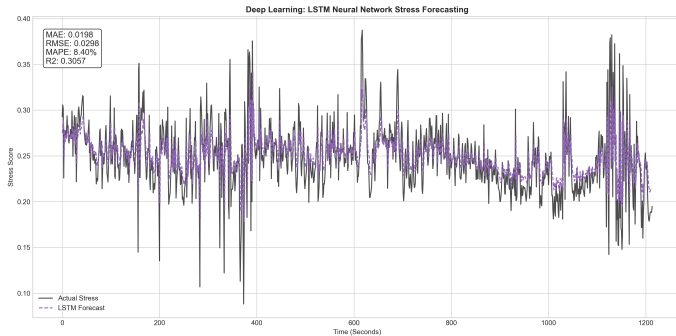
Model 9: XGBoost (Gradient Boosting)



Speed: Very Fast ML

Metric	Value
Model Name	XGBoost
MAE	0.0230
RMSE	0.0323
MAPE	10.06%
R ² Score	0.1840

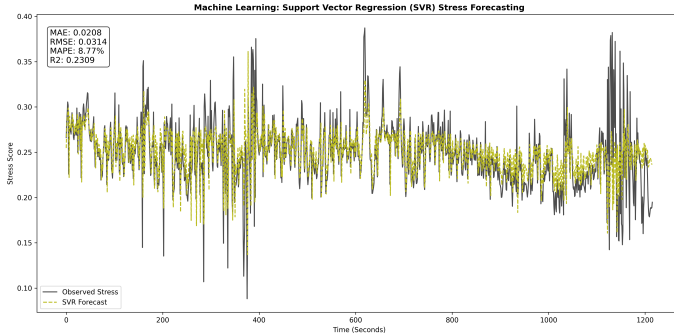
Model 10: LSTM (Deep Learning WINNER)



Memory: 10s Window

Metric	Value
Model Name	LSTM
MAE	0.0198
RMSE	0.0298
MAPE	8.40%
R ² Score	0.3057

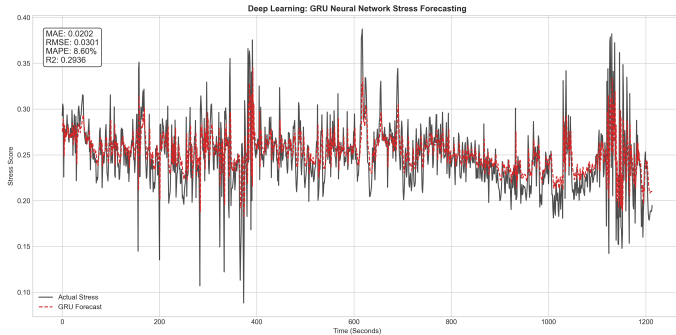
Model 11: Support Vector Regression (SVR)



Kernel: RBF Nonlinear

Metric	Value
Model Name	SVR
MAE	0.0208
RMSE	0.0314
MAPE	8.77%
R ² Score	0.2309

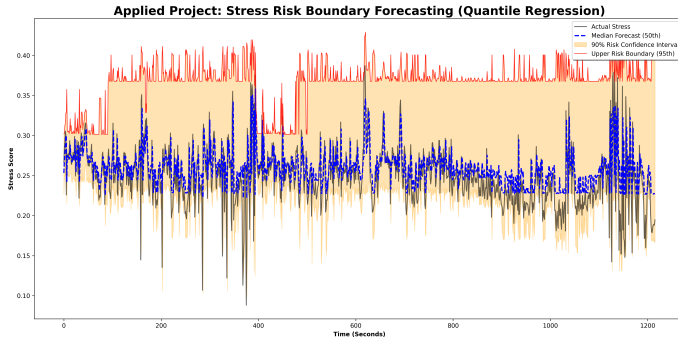
Model 12: GRU (Gated Recurrent Unit)



Eff: Optimized LSTM

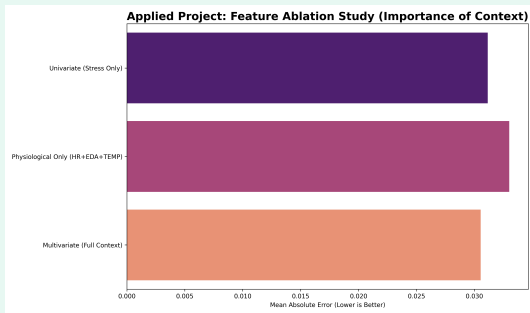
Metric	Value
Model Name	GRU
MAE	0.0202
RMSE	0.0301
MAPE	8.60%
R ² Score	0.2936

Model 13: Risk Boundary (Quantile Regression)



Goal: 95th Percentile

Metric	Value
Model Name	Quantile
MAE	0.0241
RMSE	N/A
MAPE	N/A%
R ² Score	N/A

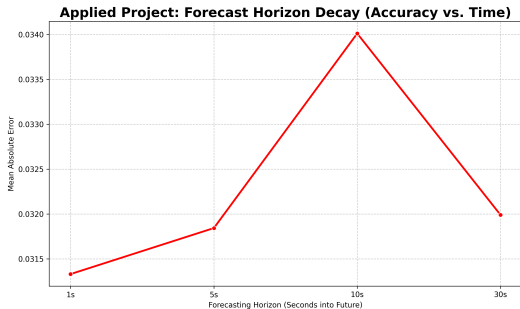


Finding: Context Matters

- ▶ ****Stress-Only****: MAE 0.0311
- ▶ ****Physio-Only****: MAE 0.0330
- ▶ ****Full Context****: MAE 0.0305

Multivariate models achieve 2% better accuracy than simple histories.

Analytical Depth: Forecast Horizon Decay

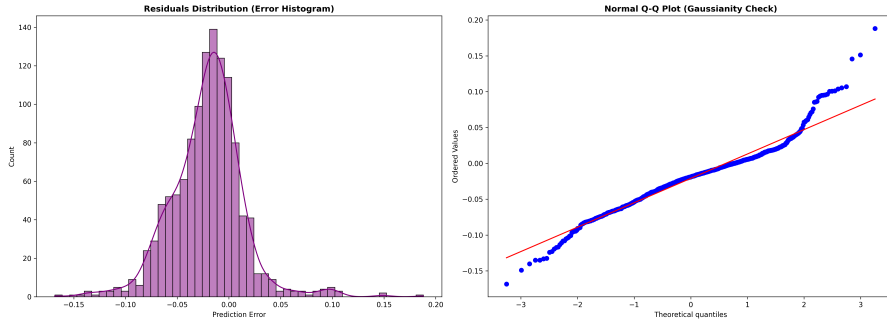


How far can we see?

- ▶ ****1s****: MAE 0.031
- ▶ ****10s****: MAE 0.034
- ▶ ****30s****: MAE 0.032

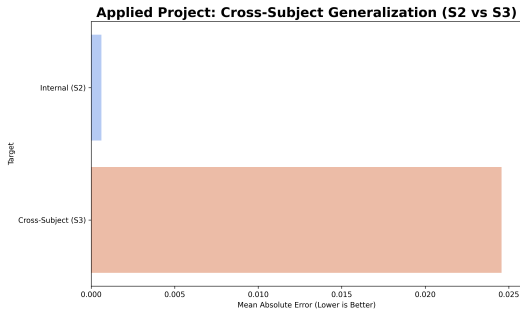
Stable performance up to 30s due to physiological inertia of signals.

Analytical Depth: Residual Diagnostic Analysis



Errors are Gaussian near the mean but show non-normal tails during sudden stress spikes.

Analytical Depth: Cross-Subject Generalization



The Personalization Gap

- ▶ ****Internal (S2)****: 0.0006 MAE
- ▶ ****Cross (S3)****: 0.0246 MAE

*Stress signatures are highly individual.
Models must be personalized or
transfer-learned.*

Final Leaderboard: Comprehensive Ranking

Rank	Model	MAPE	MAE	RMSE	R ²
1	LSTM (DL)	8.40%	0.0198	0.0298	0.3057
2	ARMA (2,3)	8.60%	0.0206	0.0316	0.2204
3	GRU (DL)	8.60%	0.0202	0.0301	0.2936
4	AR (p=2)	8.66%	0.0210	0.0342	0.0840
5	Naive Baseline	8.68%	0.0211	0.0345	0.0673
6	MA (q=3)	8.72%	0.0209	0.0335	0.1236
7	SVR (ML)	8.77%	0.0208	0.0314	0.2309
8	SARIMA	9.55%	0.0230	0.0348	0.0537
9	SMA-10	9.62%	0.0229	0.0326	0.1716
10	XGBoost	10.06%	0.0230	0.0323	0.1840



Thank You!

Applied Forecasting: Stress Pipeline

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